EXP-13 : To recover a 2 x 2 Hill key using chosen plaintext pairs.

PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
void encrypt(int x, int y, int out[2]) {
int secret[2][2] = {{9,4},{5,7}};
out[0] = (secret[0][0]*x+secret[0][1]*y)%26;
out[1] = (secret[1][0]*x+secret[1][1]*y)%26;
}
int main(void) {
int a[2], b[2];
encrypt(1,0,a); /* chosen plaintext BA */
encrypt(0,1,b); /* chosen plaintext AB */
printf("BA -> %c%c\n", a[0]+'A',a[1]+'A');
printf("AB -> %c%c\n", b[0]+'A',b[1]+'A');
printf("Recovered key:\n%d %d\n%d %d\n",
a[0],b[0],a[1],b[1]);
return 0;
}
```

OUTPUT :

```
BA -> JF
AB -> EH
Recovered key:
9 4
5 7


...Program finished with exit code 0
Press ENTER to exit console.
```